

Reusing d-DNNFs for Efficient Feature-Model Counting

SE 2026 (Originally TOSEM'24) | Chico Sundermann, Heiko Raab, Tobias Heß, Thomas Thüm, Ina Schaefer | 25.02.2026

Introduction Product Lines



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PC BUILDER

GET COMPATIBLE RECOMMENDATIONS
PICK YOUR IDEAL CORSAIR COMPONENTS

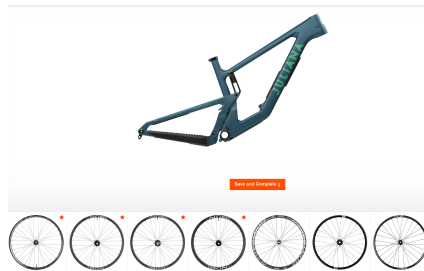


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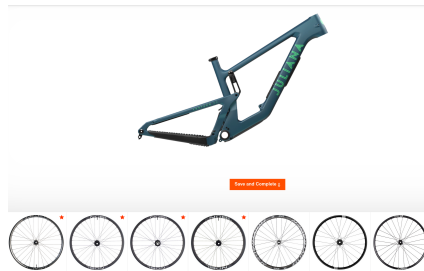
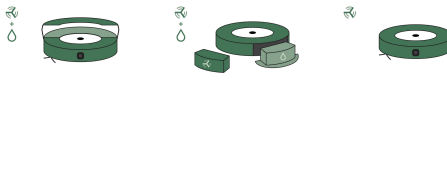


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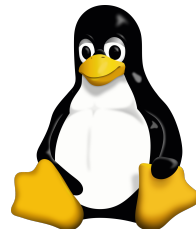
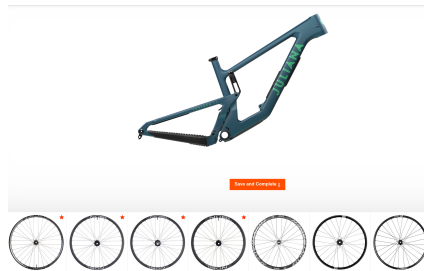


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Introduction

Feature Dependencies

 **Compatibility:** Warning! These parts have potential issues or incompatibilities. See [details](#) below.

Introduction Feature Dependencies

⚠ **Compatibility:** Warning! These parts have potential issues or incompatibilities. See [details](#) below.



Introduction

Feature Dependencies

Confirm your choice
To add the accessory you need to confirm your choice



Handlebar rear-view mirror.

Accessories to be added to the configuration:

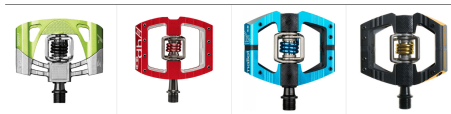
- Pair of adapters for rear-view mirror... 96881201AA Anodized ✓

Accessories to be removed from the configuration:

- Set for removing rear-view mirrors. 97381241AA Anodized ✗

Compatibility: Warning! These parts have potential issues or incompatibilities. See [details](#) below.

```
config SECURITY_INFINIBAND
bool "Infiniband Security Hooks"
depends on SECURITY && INFINIBAND
```



Introduction

Feature Dependencies

Confirm your choice

To add the accessory you need to confirm your choice



Handlebar rear-view mirror.



Accessories to be added to the configuration:



Pair of adapters for rear-view mirror...

96881201AA

Anodized



Accessories to be removed from the configuration:



Set for removing rear-view mirrors.

97381241AA

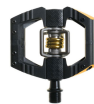
Anodized



 **Compatibility:** Warning! These parts have potential issues or incompatibilities. See [details](#) below.

```
config SECURITY_INFINIBAND
bool "Infiniband Security Hooks"
depends on SECURITY && INFINIBAND
```

 **Compatibility Warning!** CPU doesn't come with stock cooler.



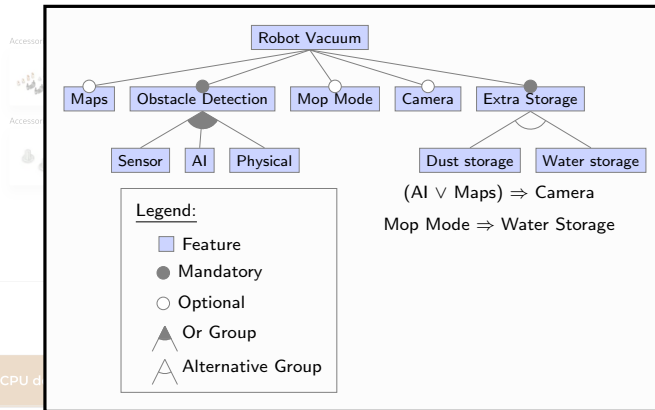
Introduction Feature Dependencies

Confirm your choice
To add the accessory you need to confirm your choice



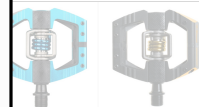
Handlebar rear-view mirror.

⚠ Compatibility Warning! CPU d

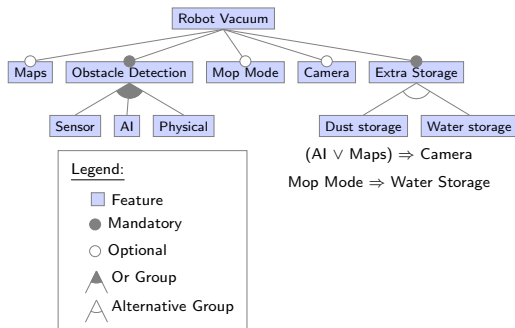


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INFINIBAND

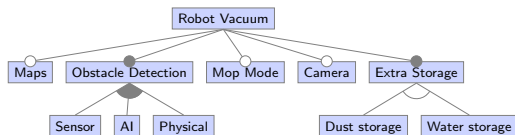


Introduction Configuration Counting



Introduction Configuration Counting

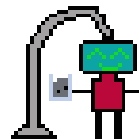
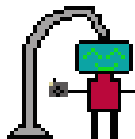
How many valid configurations?



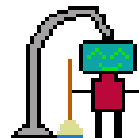
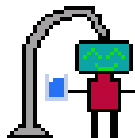
Legend:

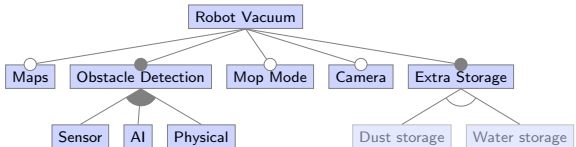
- Feature
- Mandatory
- Optional
- Or Group
- Alternative Group

$(AI \vee Maps) \Rightarrow Camera$
 $Mop Mode \Rightarrow Water Storage$



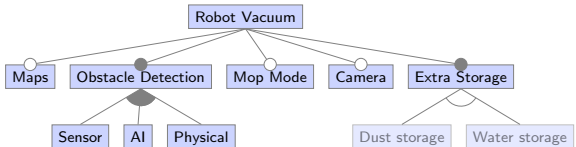
(Obviously robot vacuums)





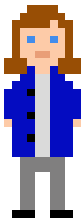
$(AI \vee Maps) \Rightarrow Camera$

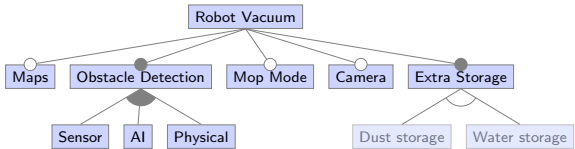
$Mop\ Mode \Rightarrow Water\ Storage$



$(AI \vee Maps) \Rightarrow Camera$
 $Mop Mode \Rightarrow Water Storage$

Extra dust or water?



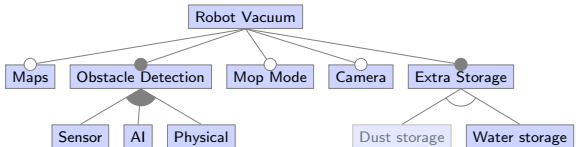


$(AI \vee Maps) \Rightarrow Camera$

$Mop\ Mode \Rightarrow Water\ Storage$

#Dust < #Water
More robots with water!



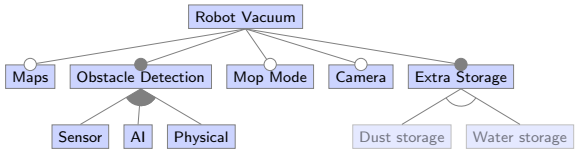


$(AI \vee Maps) \Rightarrow Camera$

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#Dust < #Water
More robots with water!





$(AI \vee Maps) \Rightarrow Camera$
 $Mop Mode \Rightarrow Water Storage$

$$\frac{\#Water}{Cost(Water)} \stackrel{?}{\lessgtr} \frac{\#Dust}{Cost(Dust)}$$





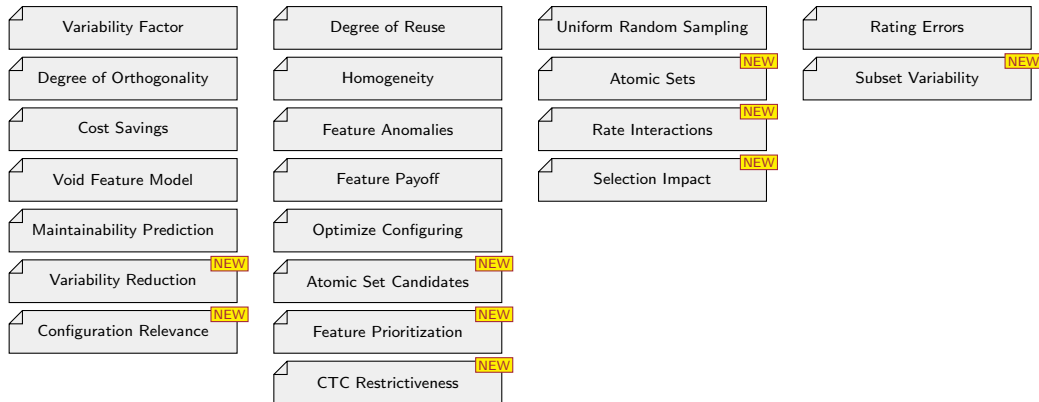
Configuration Counting Why Care?

Feature Prioritization



Configuration Counting

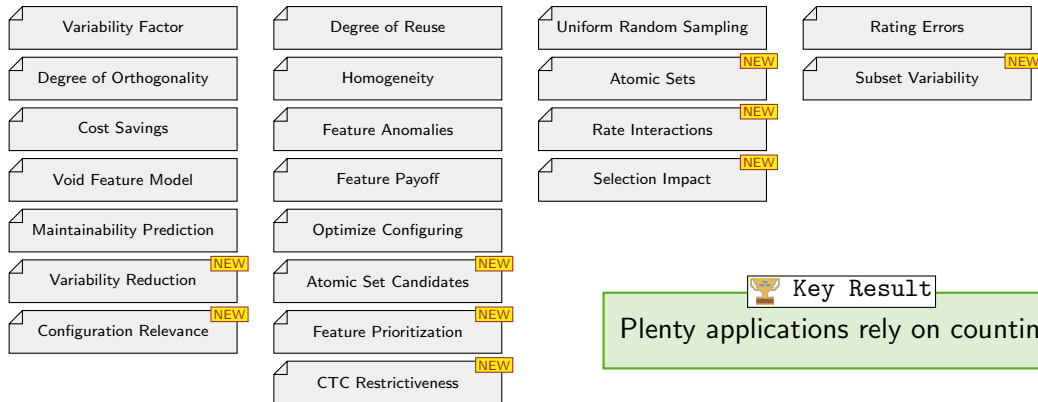
Why Care?





Configuration Counting

Why Care?



Key Result

Plenty applications rely on counting.



Configuration Counting

Why Care?

Variability Factor

Degree of Orthogonality

Cost Savings

Void Feature Model

Maintainability Prediction

Variability Reduction

Configuration Relevance

Degree of Reuse

Homogeneity

Feature Anomalies

Feature Payoff

Optimize Configuring

Atomic Set Candidates

Feature Prioritization

CTC Restrictiveness

Uniform Random Sampling

Atomic Sets

Rate Interactions

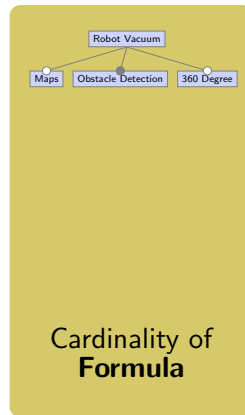
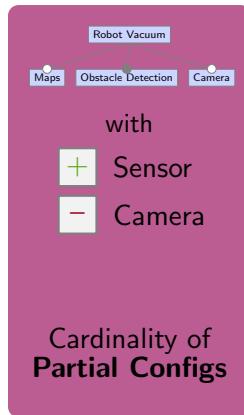
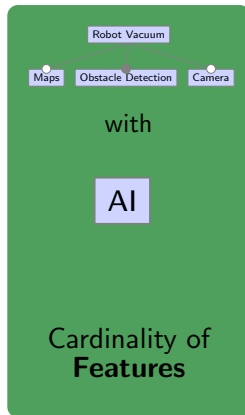
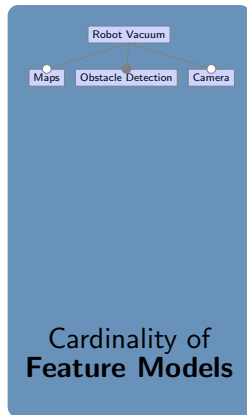
Selection Impact

Subset Variability

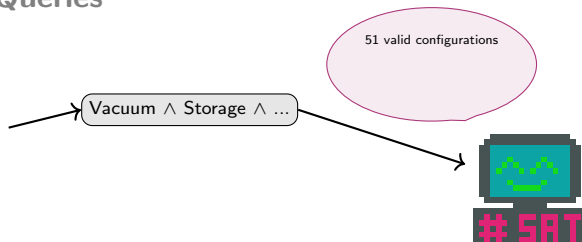
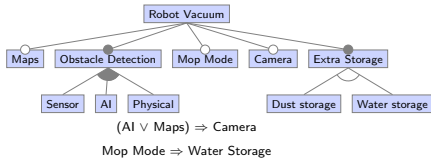
Rating Errors



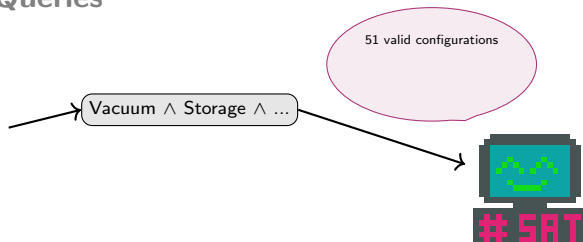
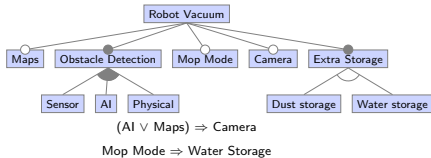
Configuration Counting Why Care?



Scalability Issues Multitude of Queries



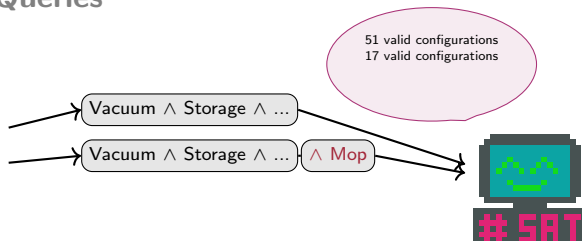
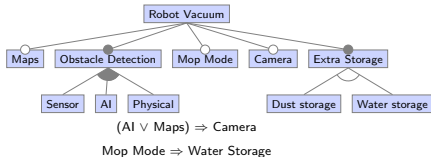
Scalability Issues Multitude of Queries



Most systems: Few seconds of runtime

EMSE'23

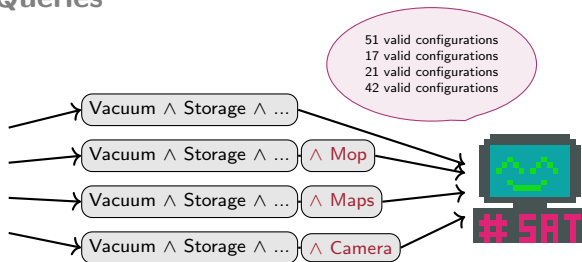
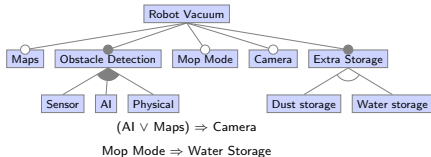
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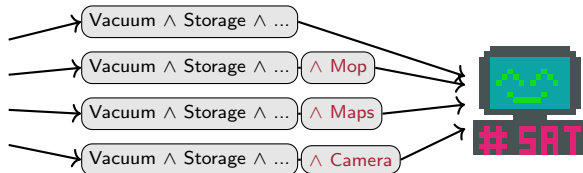
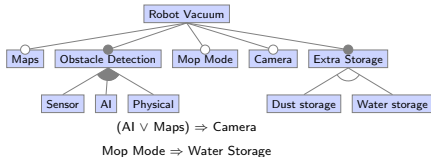
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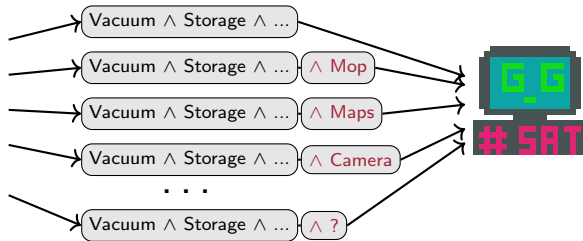
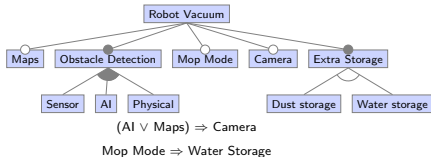
Most systems: Few seconds of runtime

EMSE'23

Type	Number of Queries	Repeatability
Features	$O(F)$	Static
	$O(F)$	Static
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	$O(F)$	Static
	$O(F)$	Static
	$O(F)$	Static
	$O(F)$	Static
	$O(F)$	Static
Partial Configurations	$O(F ^2)$	Dynamic
	$O(F ^2)$	Dynamic
	$O(F)$	Dynamic
	$O(F)$	Dynamic
	$O(F ^2)$	Dynamic
Formula	1	Dynamic
	1	Dynamic

AMAI'24

Scalability Issues Multitude of Queries



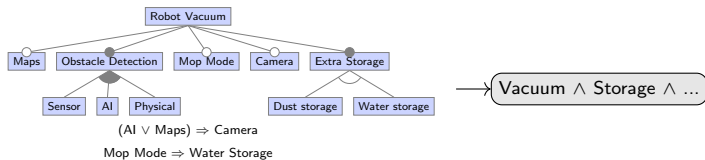
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EMSE'23

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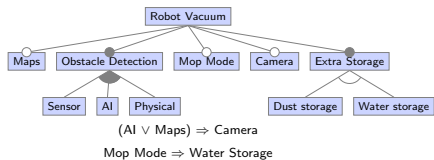
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Knowledge Compilation Concept



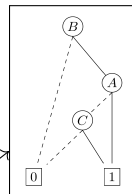
Darwiche and Marquis, A Knowledge Compilation Map, JAIR'02

Knowledge Compilation Concept

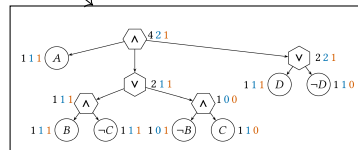


$\rightarrow Vacuum \wedge Storage \wedge \dots$

Compile

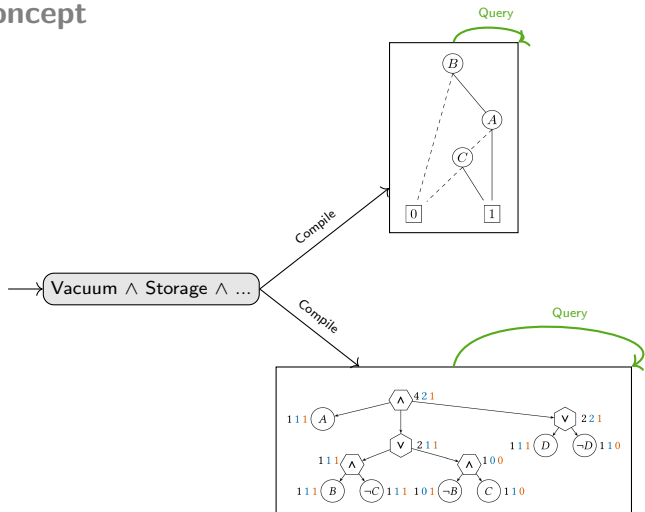
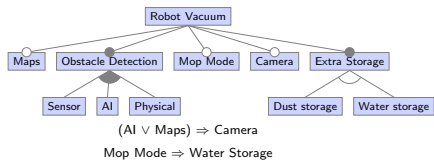


Compile



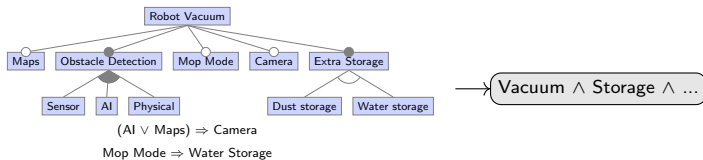
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Knowledge Compilation Concept



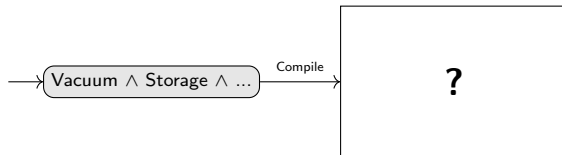
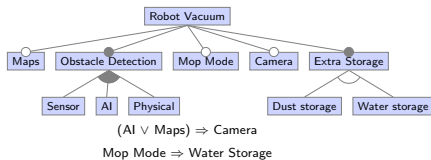
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Knowledge Compilation Which Format?



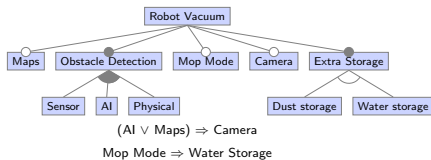
Darwiche and Marquis, A Knowledge Compilation Map, JAIR'02

Knowledge Compilation Which Format?



Darwiche and Marquis, A Knowledge Compilation Map, JAIR'02

Knowledge Compilation Which Format?



$\rightarrow Vacuum \wedge Storage \wedge \dots$ Compile $\rightarrow$

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Which format?

?

Darwiche and Marquis, A Knowledge Compilation Map, JAIR'02



Why d-DNNFs? AMAI'24

1. Which problems need to be solved?

Survey over existing analyses surveys

Classify analyses by underlying problem

2. Which knowledge compilation formats exist?

SLR on knowledge compilation

3. What are the capabilities of the formats?

Mapping languages to tractable analyses

4. How do available compilers scale?

SLR & Github search for compilers

Empirical evaluation on feature models



Why d-DNNFs? AMAI'24

1. Which problems need to be solved?

Survey over existing analyses surveys

Classify analyses by underlying problem

Which knowledge compilation format to use in **what** scenario?

3. What are the capabilities of the formats?

Mapping languages to tractable analyses

4. How do available compilers scale?

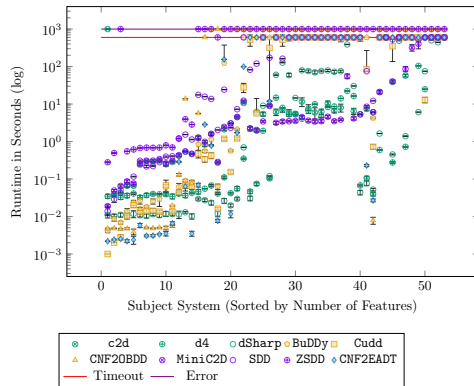
SLR & Github search for compilers

Empirical evaluation on feature models



Why d-DNNFs? Capabilities & Scalability

	#SAT-Based		
	#FM	#FM _f	#FM _c
CNF			
Horn			
EPCCL			
DNF			
DNNF			
d-DNNF	✓	✓	✓
EADT	✓	✓	✓
SDD	✓	✓	✓
PI	✓	✓	✓
IP	✓	✓	✓
ROBDD	✓	✓	✓
MODS	✓	✓	✓





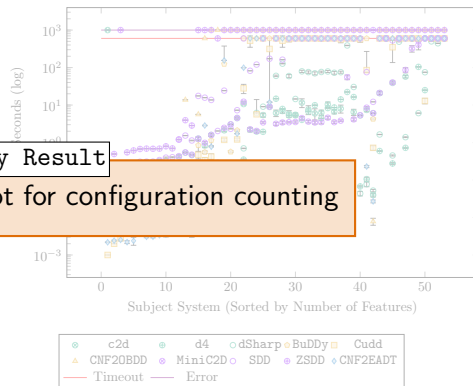
Why d-DNNFs? Capabilities & Scalability

	#SAT-Based		
	#FM	#FM _f	#FM _c
CNF			
Horn			
EPCCL			
DNF			
DNNF			
d-DNNF	✓		
EADT	✓	✓	✓
SDD	✓	✓	✓
PI	✓	✓	✓
IP	✓	✓	✓
ROBDD	✓	✓	✓
MODS	✓	✓	✓

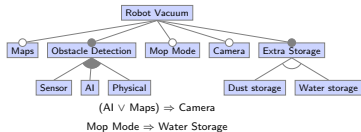


Key Result

d-DNNFs are a sweet-spot for configuration counting

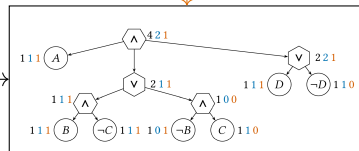


Configuration Counting with Knowledge Compilation



Vacuum $\wedge$ Storage $\wedge$...

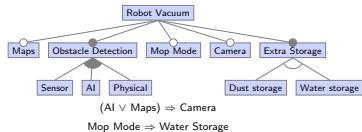
Compile



AMAI'24

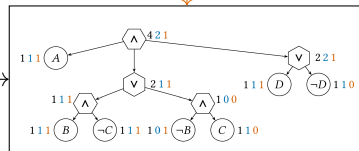
Which format?
d-DNNF

Configuration Counting with Knowledge Compilation



Vacuum $\wedge$ Storage $\wedge$...

Compile

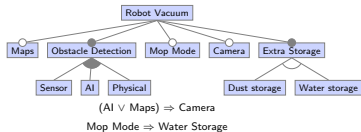


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Which format?
d-DNNF

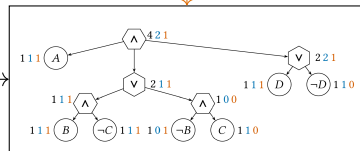
How to compile? $\rightarrow$ Off-the-shelf tools

Configuration Counting with Knowledge Compilation



Vacuum $\wedge$ Storage $\wedge$...

Compile



Query

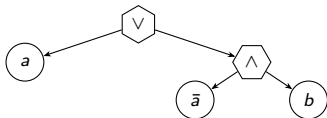
AMAI'24

Which format?
d-DNNF

TOSEM'24

How to reuse?

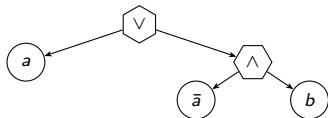
d-DNNFs Behind the Scenes



deterministic
decomposable
negation
normal
form

²Darwiche, A Compiler for Deterministic, Decomposable Negation Normal Form, AAAI'02

d-DNNFs Behind the Scenes



$\{a, b\}$
 $\{a, \bar{b}\}$

↓
2

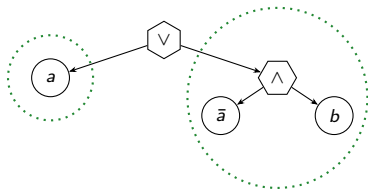
$\{\bar{a}, b\}$

↓
1

deterministic
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d-DNNFs Behind the Scenes



$\{a, b\}$
 $\{a, \bar{b}\}$ No duplicates $\{\bar{a}, b\}$

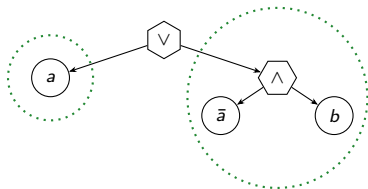
↓
2

↓
1

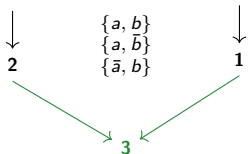
deterministic
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d-DNNFs Behind the Scenes



$\{a, b\}$
 $\{a, \bar{b}\}$ No duplicates $\{\bar{a}, b\}$



deterministic (Sum for $\vee$)

decomposable

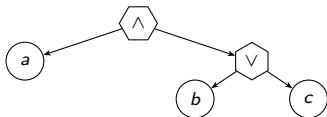
negation

normal

form

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d-DNNFs Behind the Scenes



deterministic (Sum for $\vee$)

decomposable

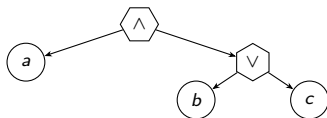
negation

normal

form

²Darwiche, A Compiler for Deterministic, Decomposable Negation Normal Form, AAAI'02

d-DNNFs Behind the Scenes



$\{a\}$

1

$\{b, c\}$
 $\{\bar{b}, c\}$
 $\{b, \bar{c}\}$

3

deterministic (Sum for $\vee$)

decomposable

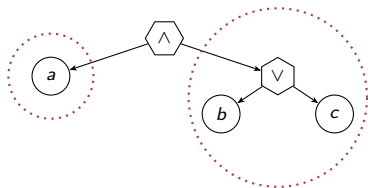
negation

normal

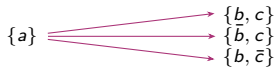
form

²Darwiche, A Compiler for Deterministic, Decomposable Negation Normal Form, AAAI'02

d-DNNFs Behind the Scenes



No shared variables



1

3

deterministic (Sum for $\vee$)

decomposable

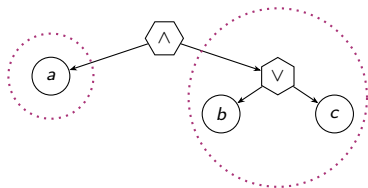
negation

normal

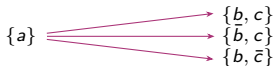
form

²Darwiche, A Compiler for Deterministic, Decomposable Negation Normal Form, AAAI'02

d-DNNFs Behind the Scenes



No shared variables



1

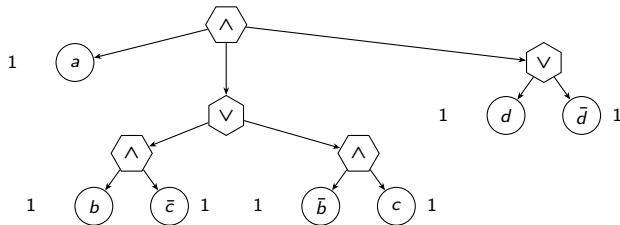
×

3

deterministic (Sum for $\vee$)
decomposable (Product for $\wedge$)
negation
normal
form

²Darwiche, A Compiler for Deterministic, Decomposable Negation Normal Form, AAAI'02

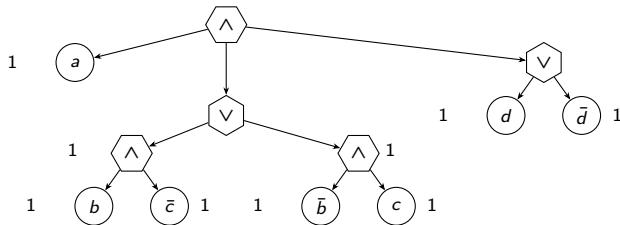
Configuration Counting with d-DNNFs



#FM: $\cdot$	$\wedge : \prod$
#f: b	$\vee : \sum$
#C _p : $\{\bar{c}, d\}$	

Basic algorithm adapted from: Darwiche, On the Tractable Counting of Theory Models and its Application to Belief Revision and Truth Maintenance, ArXiv Technical Report

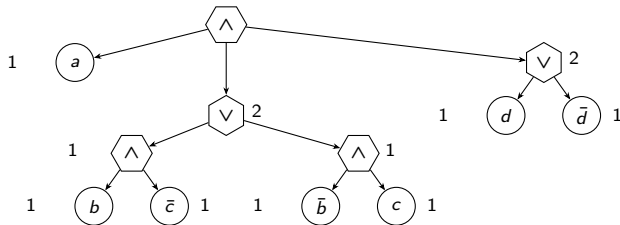
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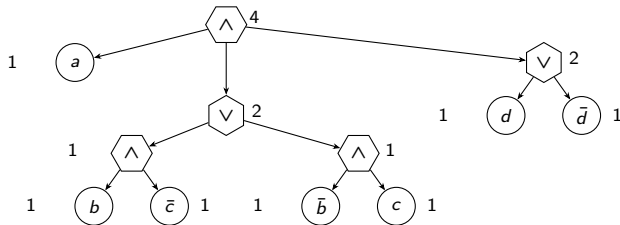
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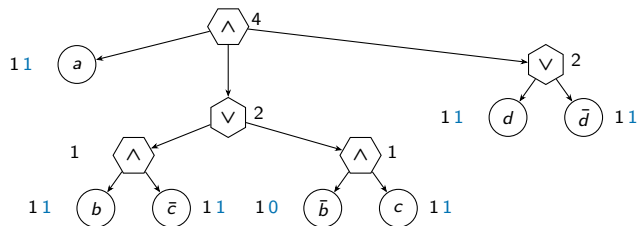
Configuration Counting with d-DNNFs



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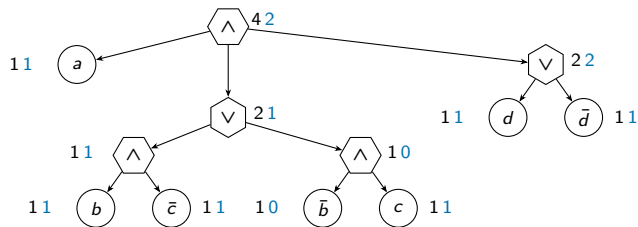
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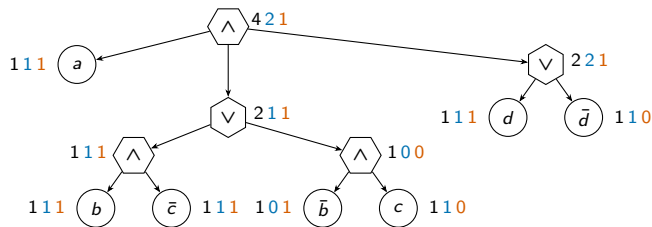
Configuration Counting with d-DNNFs



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Basic algorithm adapted from: Darwiche, On the Tractable Counting of Theory Models and its Application to Belief Revision and Truth Maintenance, ArXiv Technical Report

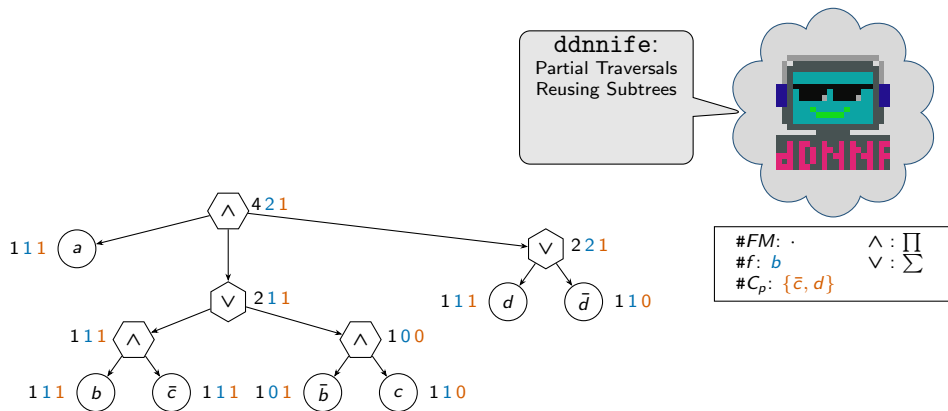
Configuration Counting with d-DNNFs



#FM: $\cdot$	$\wedge: \prod$
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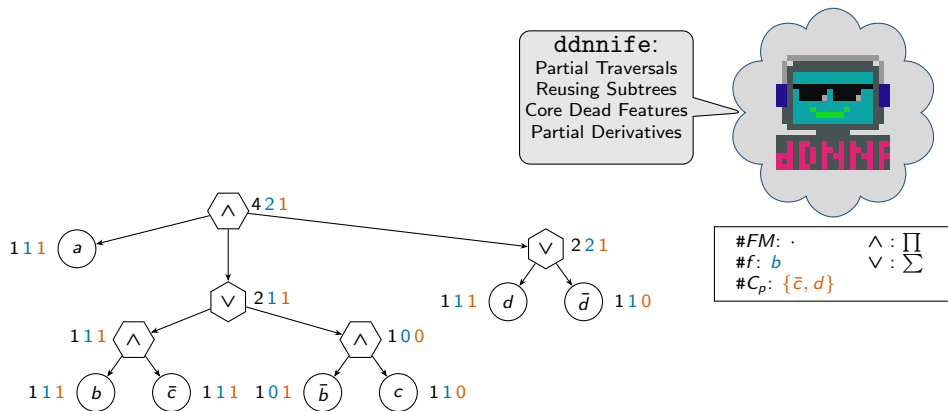
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Configuration Counting with d-DNNFs



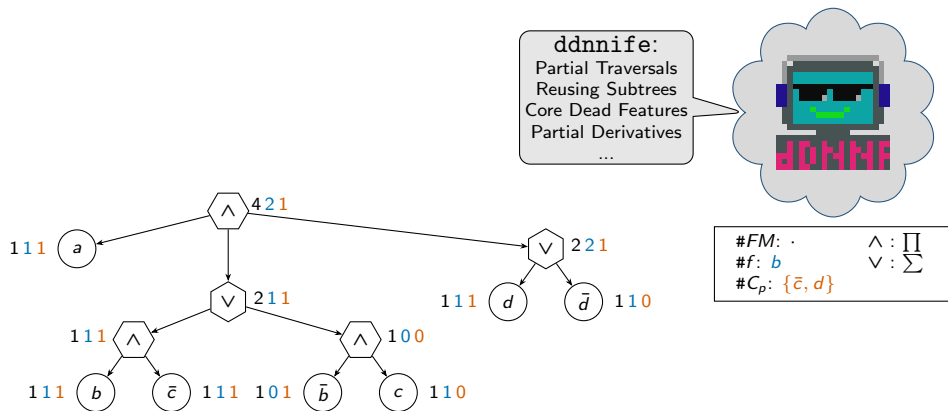
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Configuration Counting with d-DNNFs



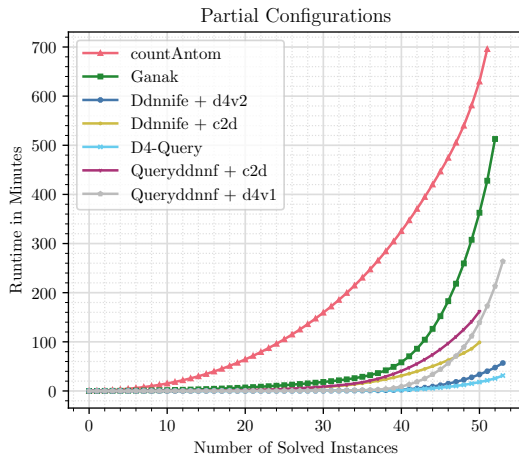
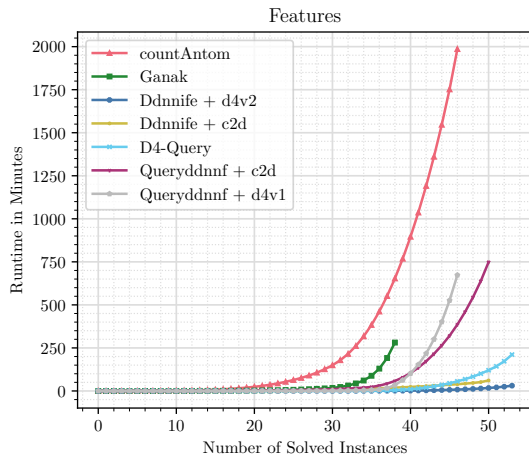
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Configuration Counting with d-DNNFs

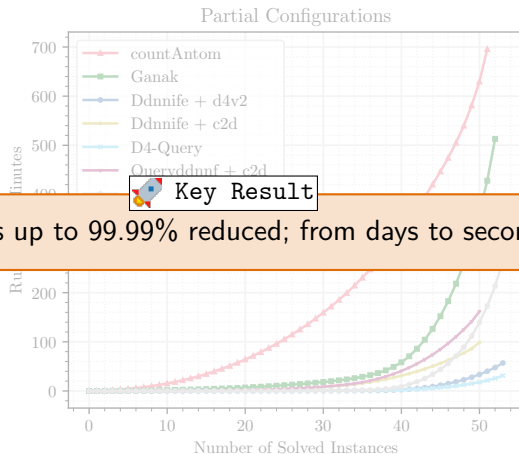
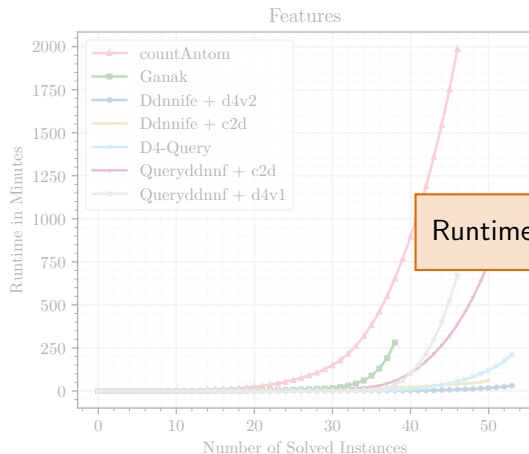


Basic algorithm adapted from: Darwiche, On the Tractable Counting of Theory Models and its Application to Belief Revision and Truth Maintenance, ArXiv Technical Report

Reusing d-DNNFs Performance



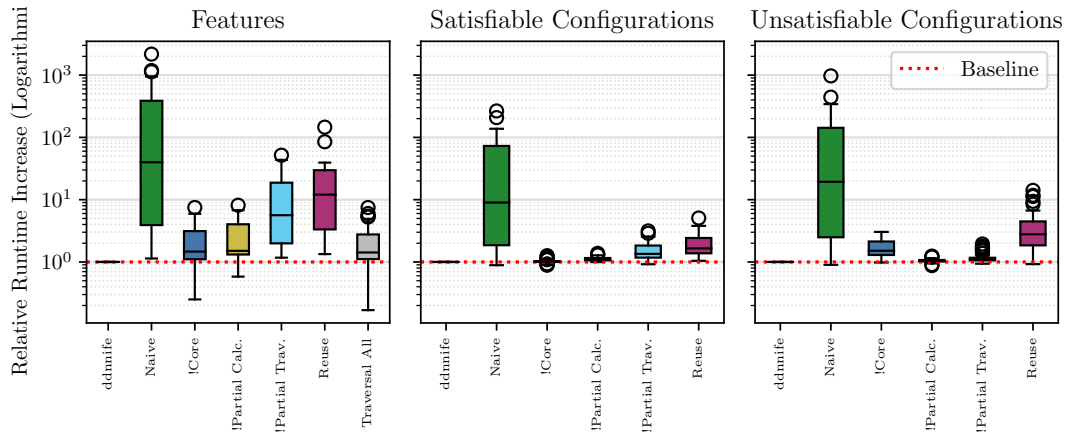
Reusing d-DNNFs Performance



Key Result

Runtimes up to 99.99% reduced; from days to seconds

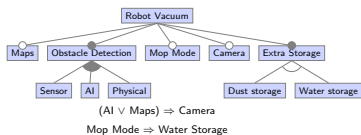
Reusing d-DNNFs Impact Optimizations





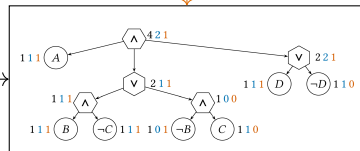
Acceleration with d-DNNF Compilation

What else?



Vacuum $\wedge$ Storage $\wedge$...

Compile



Query

AMAI'24

Which format?
d-DNNF

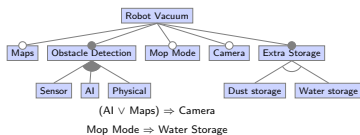
TOSEM'24

How to reuse?



Acceleration with d-DNNF Compilation

What else?

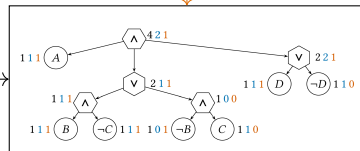


TSE Sub.

How to cope with expressive constructs?

AMAI'24

Which format?
d-DNNF



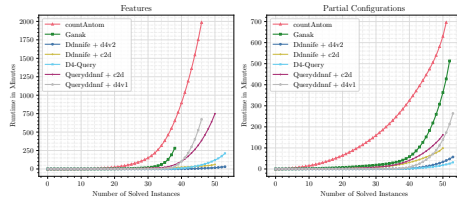
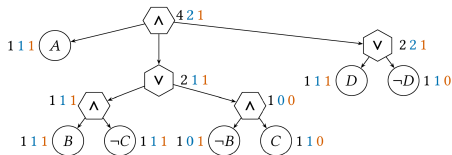
How to compile subsets?

How to reuse?

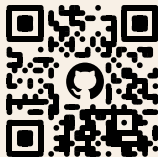
ASE'24

TOSEM'24

Reusing d-DNNFs for Efficient Feature-Model Counting

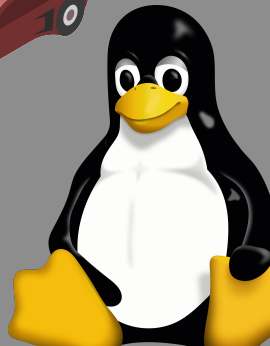
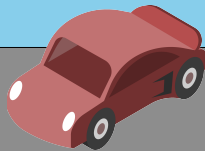


Compilation: d4 (GitHub)



Reasoning: ddnnife (GitHub)





Reusing d-DNNFs for Efficient Feature-Model Counting

SE 2026 (Originally TOSEM'24) | Chico Sundermann, Heiko Raab, Tobias Heß, Thomas Thüm, Ina Schaefer | 25.02.2026

Reusing d-DNNFs for Efficient Feature-Model Counting

1. Introduction

Product Lines

Feature Dependencies

Configuration
Counting

2. Knowledge Compilation

3. Why d-DNNFs?

AMAI'24

TOSEM'24

Content Overview